

Unit

Front-End Website Development

Title

Bean Botique Coffee Shop

Date

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Preface

Bean Boutique shop needs to have a website which is responsive, modern, user-friendly and visually responsive. By using the website, the shop can easily sell and communicate with the customers. Customers can search and buy items in anytime and anywhere.

Task 1 HTML files

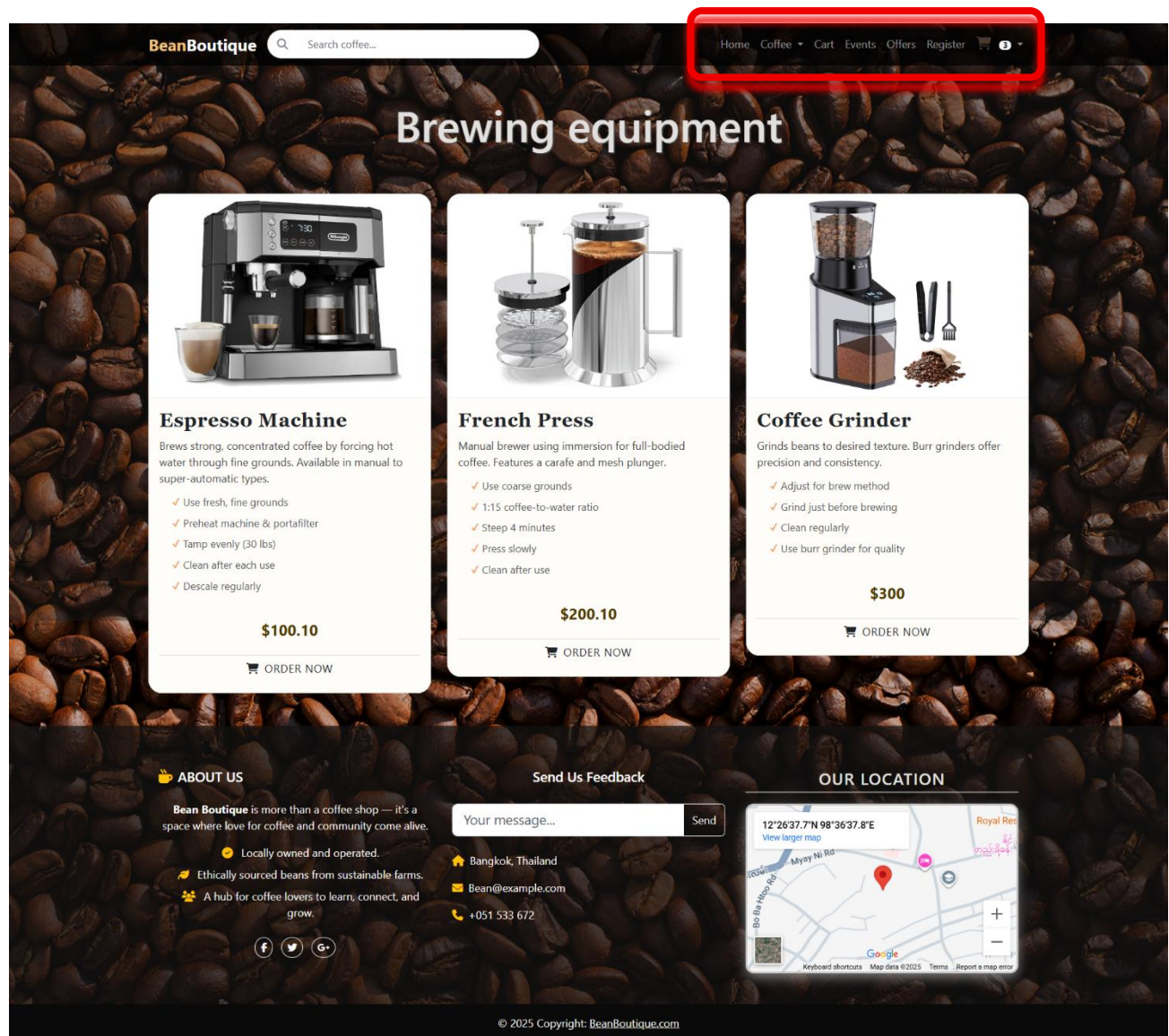


Figure 1. Six interlinked pages



Figure 2. Relevant contents

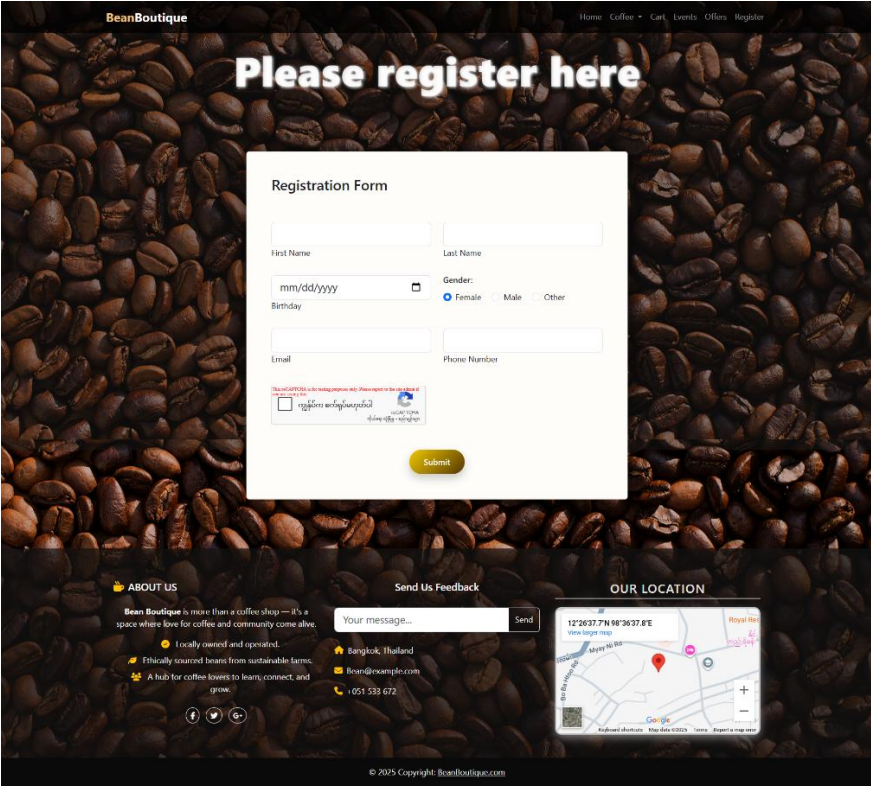


Figure 3. Footer of website

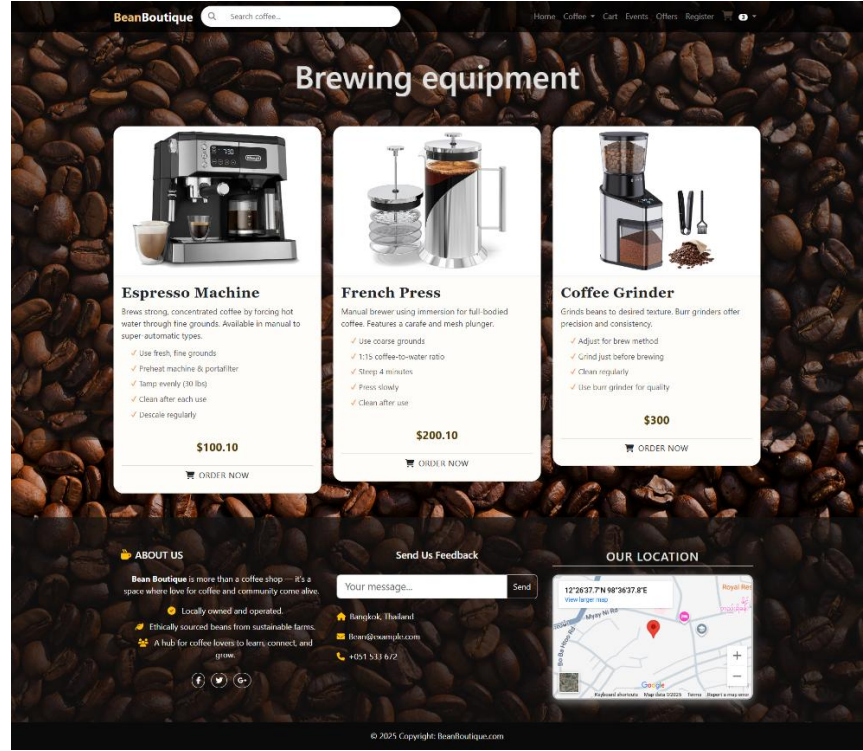


Figure 4. Brewing equipment

Task 2 External CSS file

```
# External_css.css X
C: > Users > minhe > Desktop > Assignment > FEWD > Bean Boutique Coffee Shop > # External_css.css > body
1  /* Background start*/
2  body {
3      background-image: url('Images/coffee.jpg');
4      background-size: cover;
5      background-repeat: no-repeat;
6      background-position: center center;
7      background-attachment: fixed;
8  }
9
10 body::before {
11     content: "";
12     position: fixed;
13     top: 0; left: 0;
14     width: 100%;
15     height: 100%;
16     background-color: rgba(31, 30, 30, 0.4);
17     z-index: -1;
18 }
19 /* Background end*/
20
```

Figure 5. External CSS (part -1)

```
# External_css.css X
C: > Users > minhe > Desktop > Assignment > FEWD > Bean Boutique Coffee Shop > # External_css.css > body
30
31 /* Dropdown start */
32 .To-dropdown {
33     background-color: rgba(7, 7, 7, 0.495) !important;
34     color: white;
35 }
36
37 .To-dropdown .dropdown-item {
38     color: white;
39 }
40
41 .To-dropdown .dropdown-item:hover {
42     background-color: #f0c27b;
43     color: black;
44     transition: background-color 0.3s ease, color 0.3s ease;
45 }
46 /* Dropdown end */
47
```

Figure 6. External CSS (part -2)

Task 3 Website testing

W3C validation test

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for uploaded file **index.html**

Checker Input

Show ☐ source ☐ outline ☐ image report

Check by No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

Use the Message Filtering button below to hide/show particular messages, and to see total counts of errors and warnings.

Message Filtering

1. **Error** The `center` element is obsolete. [Use CSS instead.](#)

From line 90, column 21; to line 90, column 28

```
<center><h6 c1
```

2. **Error** An `img` element must have an `alt` attribute, except under certain conditions. For details, consult [guidance on providing text alternatives for images.](#)

From line 173, column 21; to line 173, column 73

```

```

Document checking completed.

Used the HTML parser

Total execution time 7 milliseconds

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Figure 7. index.html

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for uploaded file **Select.html**

Checker Input

Show ☐ source ☐ outline ☐ image report

Check by No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

Use the Message Filtering button below to hide/show particular messages, and to see total counts of errors and warnings.

Message Filtering

1. **Error** Stray end tag `h5`.

From line 141, column 33; to line 141, column 37

```
</h5>
```

2. **Error** Attribute `invalid-attribute` not allowed on element `div` at this point.

From line 182, column 26; to line 182, column 76

```
<div class="card-body" invalid-attribute="test">
```

Attributes for element `div`:

[Global attributes](#)

Document checking completed.

Used the HTML parser

Total execution time 17 milliseconds

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Figure 8. Select.html

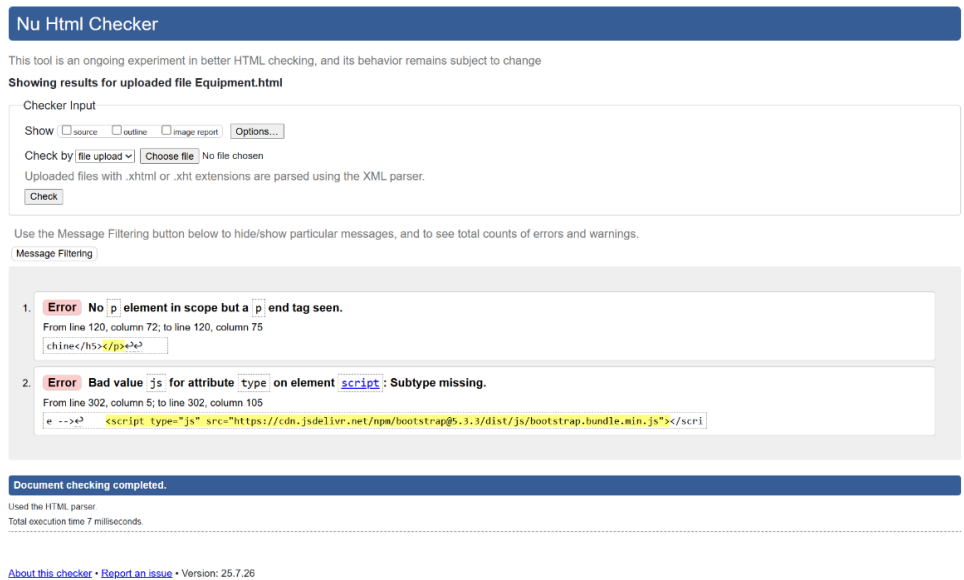


Figure 9. Equipment.html

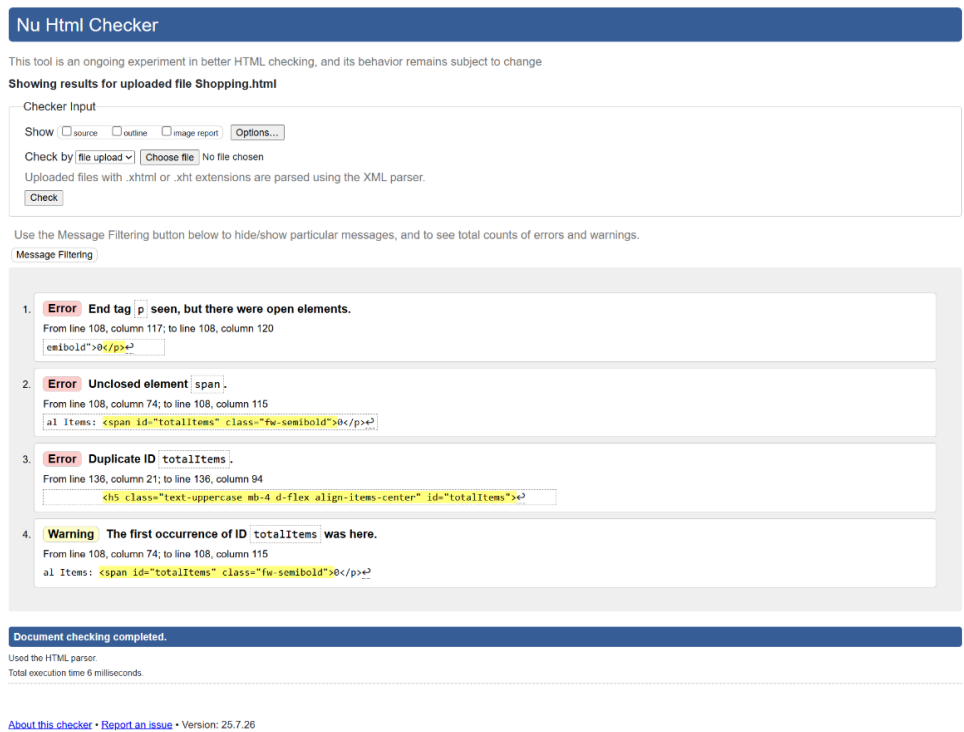


Figure 10. Shopping.html

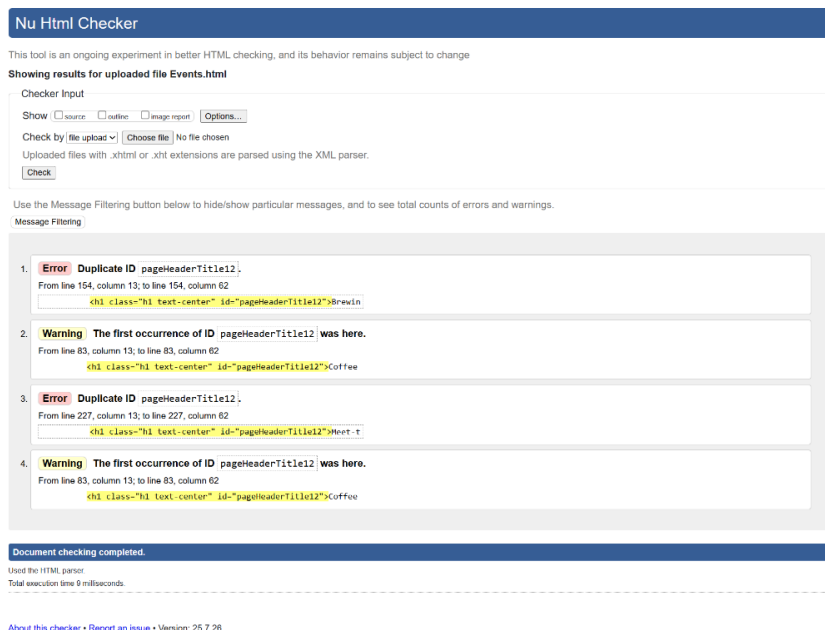


Figure 11. Events.html

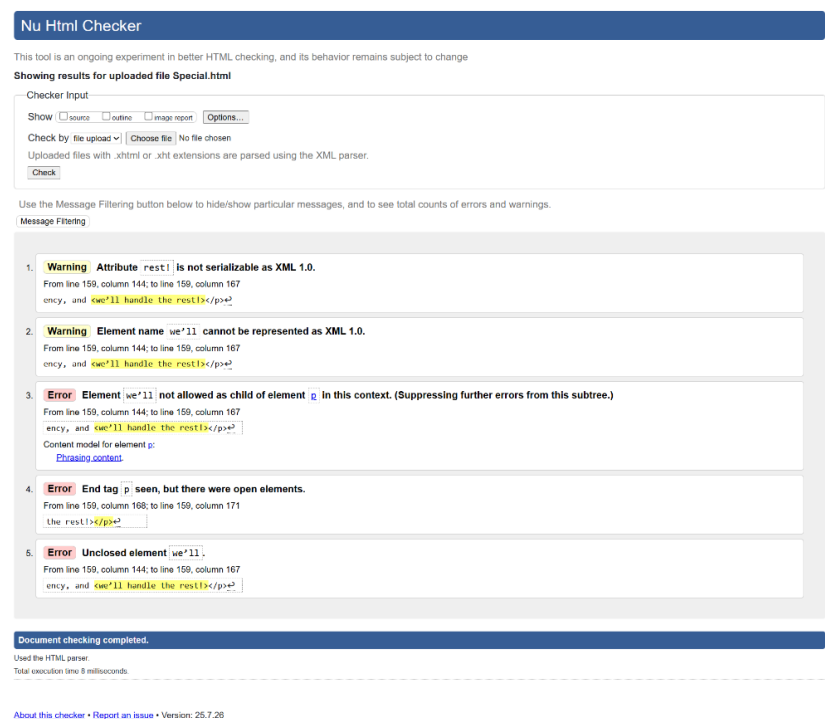


Figure 12. Special.html

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for uploaded file Register.html

Checker Input

Show ☐ source ☐ outline ☐ image report [Options...](#)

Check by [file upload](#) [Choose file](#) No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

[Check](#)

Use the Message Filtering button below to hide/show particular messages, and to see total counts of errors and warnings.

[Message Filtering](#)

1. **Error** End tag `section` seen, but there were open elements.

From line 247, column 5; to line 247, column 14

```
div>< </section>< <
```

2. **Error** Unclosed element `div`.

From line 86, column 9; to line 86, column 42

```
<div class="container py-5 h-100"><
```

Document checking completed.

Used the HTML parser.

Total execution time 7 milliseconds.

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Figure 13. Register.html

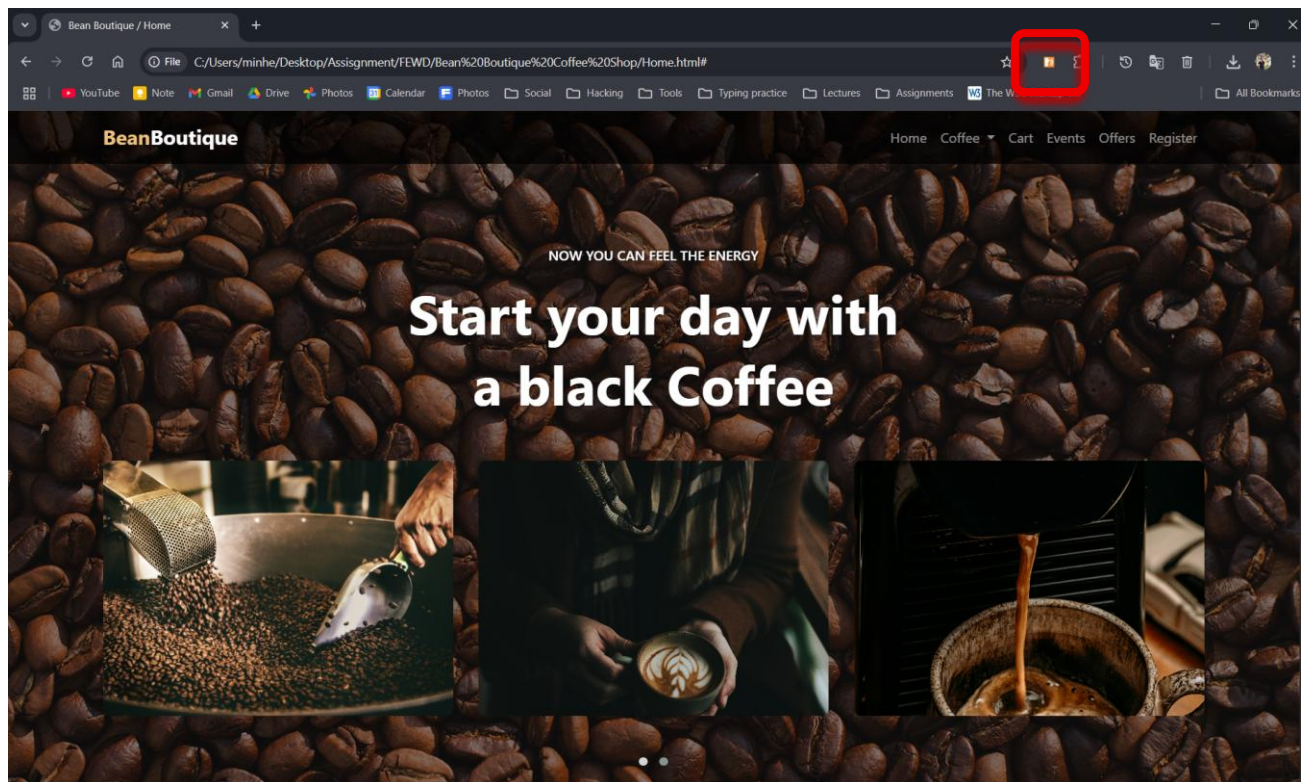


Figure 14. Screen reader

By testing screen reader in website, it can narrate loud content such as heading, paragraph and links that the pointer pointed. It can use with keyboard only by shortcut keys (e.g. tab for next focus, alt+tab for previous focus) can be used.

It can also be customized setting such as speech rate and voice volume. It supports many languages such as English, Japanese and Chinese. So, it becomes eye for disable users.

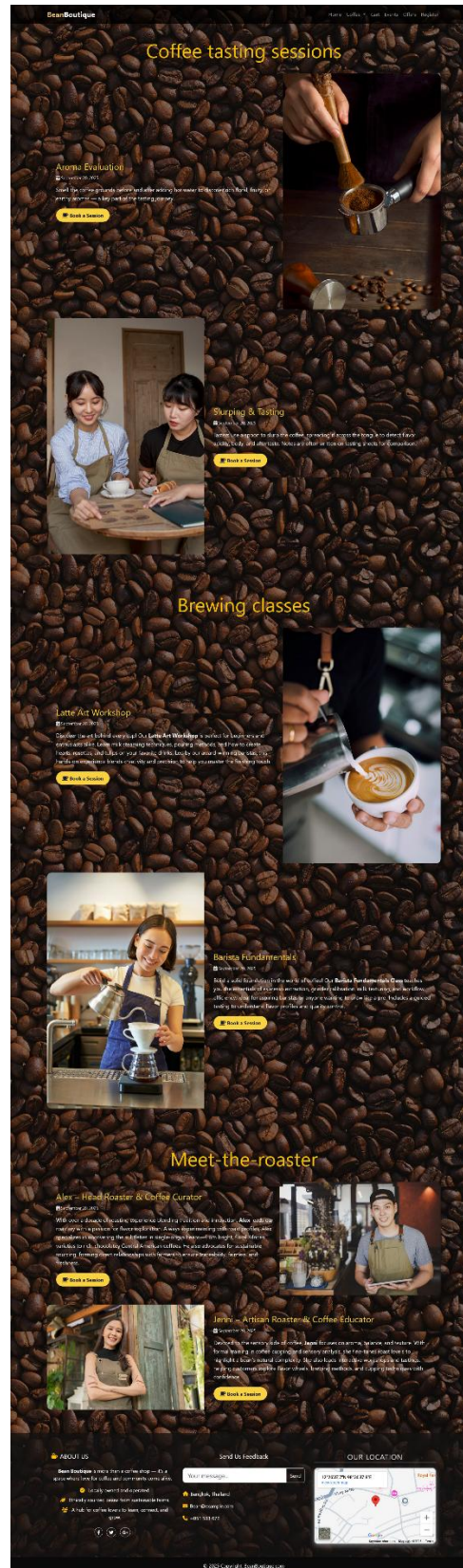


Figure 15. Test with chrome browser

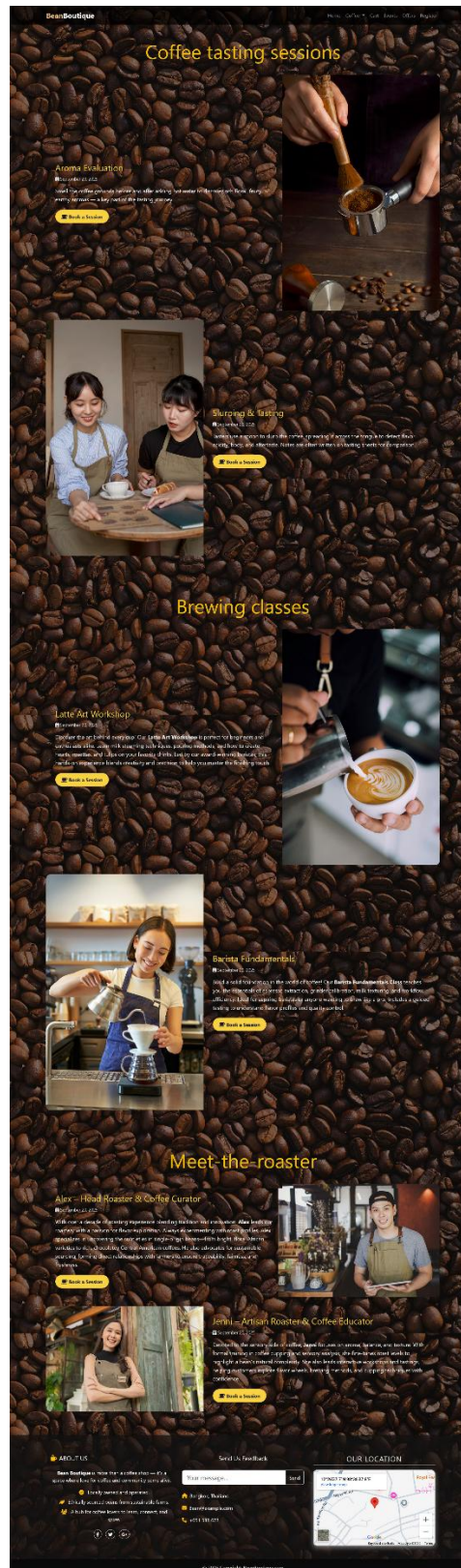


Figure 16. Test with Firefox browser

Compare browsers

In events pages, fonts in chrome are bolder than fonts in Firefox because of different text rendering engines, Chrome uses Skia and Firefox uses Cario.

In home pages, there is slightly different alignments because of slightly different calculation.



Figure 17. Chrome on mobile device

Home page

ERROR	DESCRIPTION
ERROR1	<code><center></code> element is obsolete and should use CSS instead. (line-90)
ERROR2	<code></code> element must have <code><alt></code> attribute except under certain conditions.(line-173)

For error1

`<center>` element is found at line 90. It is not necessary and it cannot make anything to heading. So, it is deleted and check heading.

For error2

The image must have appropriate contents and so `<alt>` element is added with **"Unique Coffee Blends"**.

Nu Html Checker

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Showing results for uploaded file **index.html**

Checker Input

Show ☐ source ☐ outline ☐ image report [Options...](#)

Check by file upload [Choose file](#) No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

[Check](#)

Document checking completed. No errors or warnings to show.

Used the HTML parser.

Total execution time 7 milliseconds.

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Figure 18. Corrected index.html

Coffee selection page

ERROR	DESCRIPTION
ERROR1	Stray end tag <code></h5></code> . (line-141)
ERROR2	Attribute invalid-attribute is not allowed on element <code><div></code> . (line-182)

For error1

The error is checked at line 141 and found that it is wrong end tag `</h5>`. It must be `</p>` and so replaced with `</p>` element.

For error2

The error is checked at line 182 and found that it is not necessary and also not allowed. So, **invalid-attribute** is removed and check that website is still responsive.

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for uploaded file **Select.html**

Checker Input

Show ☐ source ☐ outline ☐ image report [Options...](#)

Check by [file upload](#) No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

Document checking completed. No errors or warnings to show.

Used the HTML parser.

Total execution time 14 milliseconds.

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Figure 19. Corrected Select.html

Equipment page

ERROR	DESCRIPTION
ERROR1	No <p> element in scope but </p> end tag seen. (line-120)
ERROR2	Bad value js for attribute type on element script subtype missing. (line-302)

For error1

<p></p> element at line 120 is not needed for **Espresso Machine** heading as it already has **<h5></h5>** element. So, it is removed.

For error2

In line 302, the java script does not need **Type="js"** attribute for cart function. So it is removed and check the cart function again.

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for uploaded file **Equipment.html**

Checker Input

Show

☐ source

☐ outline

☐ image report

Options...

Check by

file upload

Choose file

No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

Check

Document checking completed. No errors or warnings to show.

Used the HTML parser.

Total execution time 10 milliseconds.

About this checker

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Figure 20. Corrected Equipment.html

Shopping cart page

ERROR	DESCRIPTION
ERROR1	End tag <p> seen, but there were open elements. (line-108)
ERROR2	No unclosed element . (line-108)
ERROR3	Duplicated id totalItems in line 136. (line-136)

For error1

Error1 occurred as a result of Error2. So it is solved after Error2 is fixed.

For error2

The error is checked at line 108 and found that **** end tag is missing. So, it is added and check the shopping cart at website.

For error3

First occurrence of id **totalItems** was found in line 108. The id in line 136 is not necessary and it is found that it has no function. So, it is removed and check the footer again.

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for uploaded file Shopping.html

Checker Input

Show

☐ source

☐ outline

☐ image report

Options...

Check by

file upload

Choose file

 No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

Check

Document checking completed. No errors or warnings to show.

Figure 21. Corrected Shopping.html

Events page

ERROR	DESCRIPTION
ERROR1	Duplicate id pageHeaderTitle12 in line 154.
ERROR2	Duplicate id pageHeaderTitle12 in line 227.

For error1

First occurrence of id **pageHeaderTitle12** was in line 83. The id **pageHeaderTitle12** in line 154 is same. So, it is changed to **pageHeaderTitle121** and create external CSS for this.

For error2

The id **pageHeaderTitle12** in line 227 is also same. So, it is changed to **pageHeaderTitle122** and also create external CSS for this.

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for uploaded file Events.html

Checker Input

Show

☐ source

☐ outline

☐ image report

Options...

Check by

file upload

Choose file

No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

Check

Document checking completed. No errors or warnings to show.

Figure 22. Corrected Event.html

Special offers page

ERROR	DESCRIPTION
ERROR1	Element we'll not allowed as child of element <p> . (line-159)
ERROR2	End tag </p> seen but there were open elements. (line-159)
ERROR3	Unclosed element we'll . (line-159)

For error1

Error is found in line 159. In passage ,**<we'll handle the rest!>** is found. **<>** is not necessary. So, it is removed and check the **coffee subscription service** card at website.

For error2 and error3

Error2 and error3 occurred as a result of Error1 and they are solved when Error1 is fixed.

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for uploaded file Special.html

Checker Input

Show

☐ source

☐ outline

☐ image report

Options...

Check by

file upload

Choose file

No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

Check

Document checking completed. No errors or warnings to show.

Used the HTML parser.

Total execution time 7 milliseconds.

About this checker

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Figure 23. Corrected Special.html

Registration page

ERROR	DESCRIPTION
ERROR1	End tag </section> seen, but there were no open elements. (line-247)
ERROR2	Unclosed element <div> in line 86.

For error1

Error1 occurred as result of Error2 and it is solved when Error2 is fixed.

For error2

Error is found at line 248. The close tag **</div>** in line 248 is missing. So, it is added and check the **registration** card again.

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for uploaded file **Register.html**

Checker Input

Show ☐ source ☐ outline ☐ image report [Options...](#)

Check by **file upload** No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 7 milliseconds.

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Figure 24. Corrected Register.html

Short report

W3c stands for world wide web consortium. It has definite standards and rules. By validation with W3c, website follows standards and so it is accessible and work well in multiple browsers.

Task 4 Website analysis

a) To get interactive and visually pleasing website

Both front-end and back-end technology are required for bean boutique website to get full functional features and services. The front end provides UI designs and performance on client side such as buttons, contents, layout and color. Back end provides behind-the-scenes works on server side such as handling requests and data from databases.

FRONT END TECHNOLOGY

HTML	for content and structure
CSS	for design and decoration
JAVA SCRIPT	for interactive features
BOOTSTRAP	Framework of HTML, CSS, Java script, used as prebuilt template
JQUERY	Java script library

So, bean boutique becomes interactivity and cross browser compatibility but it does not link with databases and so cannot get fully function. Back-end technology can be applied for communication such as storing, retrieving and editing data from databases by using PHP, Java, Ruby, Python, Nodejs framework, Mongoose library, SQL server and Oracle. Third party APIs can be used to link with other online payment and transportation systems for bean boutique.

(geeksforgeeks, 2025)

b) Uses of plugins

Three different plugins are chosen to use in bean boutique website. The reasons are as below.

1. Instagram plugin

It is social media platform with about 1 billion active users. Business can be expressed in fun and eye-catching design. Real time communication can be applied between customers and suppliers and one third of users in Instagram is looking for online purchase.

It is 10x greater brand engagement rate than Facebook and many people can be directed to bean boutique website by bio and link. It can create network with brand ambassadors and so business can get user-trust. (Kabashi, 2025)

2. Google map plugin

It is easy to use and express location for customers by various features such as turn-by-turn directions and street view. It is high accuracy and reliability with many supported languages. It is also available on both desktop and laptop and has wide users. (REDSHARK, 2025)

3. Google ReCAPTCHA plugin

It can differentiate between human users and bots and so prevents spam and malicious activity by bots. It reduces fake accounts, spam submission and fraudulent transaction. It protects both site and user personal information and also prevents slow response and downtime by reducing server load. (web999, 2024)

c) Concept and advantages of version control

Version control is used to track file changes and specific versions with time. Many developers can collaborate in same time in same codebase to improve coding efficiency. There are Local VCS, centralized VCS (e.g. SVN) and distributed VCS (e.g. Git and GitHub). Among them, Git and GitHub are common and used all over the world.

The basics of VCS

Repository	It is a storage for keeping file history and versions and can track the changes of codebase
Commit	It keeps changes with snapshots with accurate time.
Brach	It is a branch from main codebase and in each branch, developers can do coding in same time. They are in same repository and effective in developing new features and fix bugs.
Merge	It can update changes from different branches into main codebase.
Pull	Before updating into main codebase, other developer can review, discuss and approve of the changes.
Conflict	If multiple developers change data in same code file in the same time, conflict can occur. Git can show this conflict and give appropriate tools to solve them.

(Refine, 2024)

According to having these basic concepts, every code changes in GitHub can be traced and improve traceability. Many developers can work in same project at the same and so collaboration is improved. They can work in each branch with Git can merge them all. So, experiment can be done without damaging the source file and code integrity is improved. (Gravelle, 2023)

Development can restore in previous code with time and so it has also backup and redundancy. It can protect any human error as code changes can review before merging in source code. Any conflict can be solved easily. So, bean boutique website can be improved efficiently when using Git and GitHub.

d) Front end development framework

It has ready-made prebuilt reusable layouts, logics and designs to make efficiently visually appealing websites. It is also easier to develop and maintain update. Commonly used front end frameworks are Bootstrap, React, Vue.js, Angular, Svelte and Ember. They all focus on HTML, CSS and java script and suitable for large complex project.

BENEFIT OF FRONT-END DEVELOPMENT FRAMEWORK

TIME SAVING	Developers do not need to start from beginning as it has prebuilt components
CROSS-BROWSER COMPATIBILITY	It is responsive in all devices.
STRONG COMMUNITY	It is easier to find solutions and tutorials.
SCALABILITY	It supports from small to large websites.

(geeksforgeeks, 2025)

Bootstrap architecture can be used for Bean Boutique website as it is mobile first approach and responsive in all devices. Strong grid system supports complex projects. It is also time-saving as it has prebuilt components. It is easy to customized and support HTML, CSS and java script. It is open source and stable. It has strong community support and can give user-friendly and professional shopping experience. (Tim, 2019)

e) Benefit of having website

It is 24 hours online present and customers can easily find in anywhere and anytime. It can provide all information such as advertising about Bean Boutique in one place and effective in information exchange with customer's feedback. It is also cost saving and easy to expand business. Brand loyalty can build and also analysis customer behaviors. (Parachute, 2025)

To improve the website development, following technologies can be added.

TECHNOLOGIES

BACK-END TECHNOLOGY	It is used to communicate with database and get fully function.
ARTIFICIAL INTELLIGENCE	AI can be used in customer service for 24-hour availability.
DARK MODE	Many users enjoy it and it can save user's device energy.
PWA FEATURES	The website can be used as mobile app with offline access and added to home screen.

Conclusion

Bean Boutique website is now built only with front-end technology and should link with back-end technology to get the well-functional website. Many additional features can be added to expand the business and users' satisfaction can be increased.

-----End of the Assignment-----

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